



Fig. 2: Pin diagram of Winkler Board

The secret

By bringing together all basic components that may be used for an IoT project and interfacing these to a powerful ATmega core, Winkler Board was created with the aim of being not just compact but also cheap.

The new powerful microcontroller core. The core of Winkler Board is Atmel ATmega128 microcontroller, which is accompanied by onboard Wi-Fi controller, radio transceiver, Bluetooth Low Energy, real-time clock, accelerometer, gyroscope and more. It is an all-in-one module that provides all basic peripherals needed for a prototyping idea.

The board is equipped with 38 digital input pins, seven pulse-width modulation digital I/O pins and eight analogue input pins. It also has a connector input for power and programming port wired micro-USB, thereby facilitating numerous operations.

Use only what you need. On-board peripherals are built with Smart Opt, which helps you power down the components that are of no use to the project. This selective turn-off feature is a big boon for battery-powered devices for saving power.

Code travels over the air. Winkler supports over-the-air programming, which helps you burn the code to ATmega128 and ESP8266 wirelessly. "This is accomplished by pairing the board's HC-05 Bluetooth module to a laptop and then using Bluetooth's serial COM port to upload the new program," says Hedge.

Dual microcontroller units. ESP8266 being a Wi-Fi module with a system-on-chip, ATmega128 and ESP8266 ESP12E combination on Winkler Board provides dual microcontrollers. The two universal asynchronous receivers/transmitters of ATmega128 are connected to ESP8266, thereby helping these communicate with each other at the same time these run their own code.

Ease of programming. Winkler Board can be programmed with Arduino IDE via a micro-USB cable, but you could also program ATmega128 over Bluetooth and ESP12 module over Wi-Fi wirelessly. This makes it easy to update the firmware while the board is inside a case or is hard to access. Winkler Board also supports conventional wired serial programming.

Open source hardware platform

The best part of Winkler Board is that it is completely open source. The

TECHNICAL SPECIFICATIONS

Specifications	Value
Microcontroller	ATmega128
Output voltage	5V
Digital I/O pins	38
PWM digital I/O pins	7
Analogue input pins	8
Flash memory	128kB
SRAM	4kB
EEPROM	16MHz
Onboard Wi-Fi	ESP8266-ESP12E
Onboard Bluetooth	HC-05
Onboard radio	NRF2401
Real-time clock	DS3231

source code and the pin diagram are available at [Github](#), [Instructables](#), [crowdsupply.com](#) and other forums. This is a huge contribution to the open source hardware community, which facilitates makers like you to make your own Winkler Board.

Winkler Board is truly a boon for makers who work on electronic projects that involve a lot of I/O operations, and involve using different communication protocols to fulfil different tasks. It gives you more time to focus on actual building and completing the important parts of your project rather than wasting it on routine, common-place, low-level tasks. **EFY**